

DSA Master Sheet

Topic 1 — Arrays

40 Problems	14 Easy	18 Medium	8 Hard
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Solve in order: Easy → Medium → Hard. Every problem has a direct LeetCode link.
Language: Java | Goal: FAANG + Internship + College exams

Golden Tips — Follow These for Every Problem

1. Struggle before you search

Spend at least 20 minutes on every problem before looking at hints or solutions. The struggle is where logic and pattern recognition actually get built.

2. Brute force first, then optimize

Always write the brute force solution first. Get it working, then improve. $O(n^2)$ that works beats $O(n)$ that doesn't.

3. Write the pattern after every problem

In your notebook, write one line: 'This was a [pattern] problem because [reason].' This habit builds interview instinct faster than anything else.

4. Revise after 7 days

Re-solve every problem from scratch 7 days later without looking at your previous code. If you can't, you haven't actually learned it yet.

5. Always state Big O before submitting

Before you click Submit, write the time and space complexity in a comment. Interviewers always ask. Make it a non-negotiable habit now.

6. Test edge cases manually

Before running, dry-run your code on: empty array, single element, all same elements, negative numbers. Catch bugs yourself.

Easy — Build Your Base

1. Contains Duplicate

Pattern: Hashing

<https://leetcode.com/problems/contains-duplicate/>

Tip: Use HashSet. Add each element; if it already exists, return true. $O(n)$ time, $O(n)$ space.

2. Two Sum

Pattern: Hashing

<https://leetcode.com/problems/two-sum/>

Tip: Store each number's index in a HashMap. For each element, check if target-num exists.

3. Best Time to Buy and Sell Stock

Pattern: Greedy / Sliding Window

<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/>

Tip: Track minimum price seen so far. At each step, profit = current - min. Track max profit.

4. Missing Number	Pattern: Math / XOR
https://leetcode.com/problems/missing-number/	
<i>Tip: Expected sum = $n*(n+1)/2$. Subtract actual sum. Or use XOR of 0..n XOR all elements.</i>	
5. Find All Numbers Disappeared in Array	Pattern: Index Marking
https://leetcode.com/problems/find-all-numbers-disappeared-in-an-array/	
<i>Tip: Mark visited indices by negating $arr[abs(arr[i])-1]$. Indices still positive at end are missing.</i>	
6. Single Number	Pattern: XOR Bit Trick
https://leetcode.com/problems/single-number/	
<i>Tip: XOR all elements. Duplicate numbers cancel out (a XOR a = 0). Result is the single number.</i>	
7. Move Zeroes	Pattern: Two Pointers
https://leetcode.com/problems/move-zeroes/	
<i>Tip: Use a slow pointer for next non-zero position. Swap non-zero elements forward in one pass.</i>	
8. Majority Element	Pattern: Boyer-Moore Voting
https://leetcode.com/problems/majority-element/	
<i>Tip: Candidate + count approach. Count goes up for same element, down for different. $O(1)$ space.</i>	
9. Running Sum of 1D Array	Pattern: Prefix Sum
https://leetcode.com/problems/running-sum-of-1d-array/	
<i>Tip: $arr[i] += arr[i-1]$. Classic prefix sum pattern. Foundation for many harder problems.</i>	
10. Shuffle the Array	Pattern: Index Manipulation
https://leetcode.com/problems/shuffle-the-array/	
<i>Tip: Place $nums[i]$ and $nums[i+n]$ alternately. Good practice for index arithmetic.</i>	
11. Intersection of Two Arrays II	Pattern: Hashing / Two Pointers
https://leetcode.com/problems/intersection-of-two-arrays-ii/	
<i>Tip: HashMap to count freq of $arr1$. For each element in $arr2$, if $count > 0$ add to result.</i>	
12. Maximum Average Subarray I	Pattern: Fixed Sliding Window
https://leetcode.com/problems/maximum-average-subarray-i/	
<i>Tip: Compute first window sum. Slide by adding next element and removing first. Track max sum.</i>	
13. Squares of a Sorted Array	Pattern: Two Pointers
https://leetcode.com/problems/squares-of-a-sorted-array/	
<i>Tip: Use left and right pointers. Compare absolute values, fill result array from the back.</i>	
14. Plus One	Pattern: Array Simulation
https://leetcode.com/problems/plus-one/	
<i>Tip: Traverse from end. Add 1, carry if 9. If all 9s, create new array with 1 prepended.</i>	

15. Maximum Subarray (Kadane's Algorithm)	Pattern: Kadane's Algorithm
https://leetcode.com/problems/maximum-subarray/	
<i>Tip: $currentSum = \max(num, currentSum + num)$. $maxSum = \max(maxSum, currentSum)$. Must memorize.</i>	
16. Sort Colors (Sort 0s 1s 2s)	Pattern: Dutch National Flag / 3 Pointers
https://leetcode.com/problems/sort-colors/	
<i>Tip: Three pointers: $low=0$, $mid=0$, $high=n-1$. Swap based on value. $O(n)$ time, $O(1)$ space.</i>	
17. Product of Array Except Self	Pattern: Prefix + Suffix Product
https://leetcode.com/problems/product-of-array-except-self/	
<i>Tip: Left pass: prefix products. Right pass: multiply suffix products. No division needed.</i>	
18. 3Sum	Pattern: Sort + Two Pointers
https://leetcode.com/problems/3sum/	
<i>Tip: Sort array. Fix one element, use two pointers on the rest. Skip duplicates carefully.</i>	
19. Container With Most Water	Pattern: Two Pointers
https://leetcode.com/problems/container-with-most-water/	
<i>Tip: Left and right pointers. $Area = \min(height[l], height[r]) * (r-l)$. Move the shorter side.</i>	
20. Subarray Sum Equals K	Pattern: Prefix Sum + HashMap
https://leetcode.com/problems/subarray-sum-equals-k/	
<i>Tip: HashMap stores prefix sum frequencies. If $prefixSum - k$ exists in map, add its count.</i>	
21. Find the Duplicate Number	Pattern: Floyd's Cycle Detection
https://leetcode.com/problems/find-the-duplicate-number/	
<i>Tip: Treat array as linked list. Fast and slow pointer find the cycle entry = duplicate.</i>	
22. Search in Rotated Sorted Array	Pattern: Modified Binary Search
https://leetcode.com/problems/search-in-rotated-sorted-array/	
<i>Tip: Check which half is sorted. Determine if target is in sorted half, then binary search it.</i>	
23. Find Minimum in Rotated Sorted Array	Pattern: Modified Binary Search
https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/	
<i>Tip: If $mid > right$, minimum is in right half. Else in left half. $O(\log n)$.</i>	
24. Longest Consecutive Sequence	Pattern: HashSet
https://leetcode.com/problems/longest-consecutive-sequence/	
<i>Tip: Add all to HashSet. For each n, check if $n-1$ exists (start of sequence). Count forward.</i>	
25. Rotate Array	Pattern: Reverse Technique
https://leetcode.com/problems/rotate-array/	
<i>Tip: Reverse all, reverse first k, reverse last $n-k$. $O(n)$ time, $O(1)$ space. Elegant trick.</i>	

26. 4Sum	Pattern: Sort + Two Pointers
https://leetcode.com/problems/4sum/	
<i>Tip: Two nested loops + two pointers. Extension of 3Sum. Handle duplicates at each level.</i>	
27. Maximum Product Subarray	Pattern: DP / Kadane's Variant
https://leetcode.com/problems/maximum-product-subarray/	
<i>Tip: Track both max and min product (negatives flip sign). $maxProd = \max(num, num * max, num * min)$.</i>	
28. Spiral Matrix	Pattern: Simulation / Boundary Shrink
https://leetcode.com/problems/spiral-matrix/	
<i>Tip: Use top, bottom, left, right boundaries. Traverse and shrink. Go right, down, left, up.</i>	
29. Set Matrix Zeroes	Pattern: In-place Marking
https://leetcode.com/problems/set-matrix-zeroes/	
<i>Tip: Use first row/col as markers. First pass: mark. Second pass: zero out. Handle row/col 0 separately.</i>	
30. Jump Game	Pattern: Greedy
https://leetcode.com/problems/jump-game/	
<i>Tip: Track $maxReach = \max(maxReach, i + nums[i])$. If $i > maxReach$ at any point, return false.</i>	
31. Merge Intervals	Pattern: Sort + Merge
https://leetcode.com/problems/merge-intervals/	
<i>Tip: Sort by start. If current start $\leq$ prev end, merge (update end). Else add to result.</i>	
32. Next Permutation	Pattern: Array Manipulation
https://leetcode.com/problems/next-permutation/	
<i>Tip: Find rightmost ascent, swap with next greater on right, reverse the suffix. $O(n)$ time.</i>	

Hard — Crack Any Interview

33. Trapping Rain Water	Pattern: Two Pointers
https://leetcode.com/problems/trapping-rain-water/	
<i>Tip: Left and right pointers. Water at $i = \min(maxLeft, maxRight) - height[i]$. Move shorter side.</i>	
34. Jump Game II	Pattern: Greedy BFS
https://leetcode.com/problems/jump-game-ii/	
<i>Tip: Think in levels like BFS. Track current range end and farthest reachable. Count level jumps.</i>	
35. First Missing Positive	Pattern: Cyclic Sort / Index Marking
https://leetcode.com/problems/first-missing-positive/	
<i>Tip: Place each positive num i at index $i-1$. Then find first index where $arr[i] \neq i+1$.</i>	
36. Largest Rectangle in Histogram	Pattern: Monotonic Stack
https://leetcode.com/problems/largest-rectangle-in-histogram/	
<i>Tip: Stack stores increasing bar indices. Pop when shorter bar found. Area = height * width span.</i>	

37. Maximum Gap	Pattern: Bucket Sort
https://leetcode.com/problems/maximum-gap/	
<i>Tip: Pigeonhole: $\max \text{ gap} \geq (\max - \min) / (n - 1)$. Use $n - 1$ buckets. Max gap is between buckets.</i>	
38. Count of Smaller Numbers After Self	Pattern: Merge Sort / BIT
https://leetcode.com/problems/count-of-smaller-numbers-after-self/	
<i>Tip: Use modified merge sort. Count inversions during merge. Fenwick tree is the $O(n \log n)$ approach.</i>	
39. Median of Two Sorted Arrays	Pattern: Binary Search on Partition
https://leetcode.com/problems/median-of-two-sorted-arrays/	
<i>Tip: Binary search on smaller array. Find partition where left halves $\leq$ right halves. $O(\log(\min(m, n)))$.</i>	
40. Best Time to Buy and Sell Stock III	Pattern: DP / State Machine
https://leetcode.com/problems/best-time-to-buy-and-sell-stock-iii/	
<i>Tip: 4 states: buy1, sell1, buy2, sell2. Track max profit for each state across all days.</i>	

Finish all 40 problems → ask Claude for Topic 2: Strings (25 problems) with links
 DSA Master Sheet by Claude | Topic 1 of 12 | Java | FAANG + Internship Prep